Cranberries and cranberry products: powerful in vitro, ex vivo, and in vivo sources of antioxidants.

Cranberry products and especially cranberry juice (CJ) have been consumed for health reasons primarily due to their effect on urinary tract infections. We investigated the quantity of both free and total (after hydrolysis) phenolic antioxidants in cranberry products using the Folin assay. The order of amount of total polyphenols in cranberry foods on a fresh weight basis was as follows: dried > frozen > sauce > jellied sauce. On a serving size basis for all cranberry products, the order was as follows: frozen > 100% juice > dried > 27% juice > sauce > jellied sauce. High fructose corn syrup (HFCS) is a major source of sugar consumption in the U.S. and contains both glucose and fructose, potential mediators of oxidative stress. We investigated the effect of the consumption of HFCS and ascorbate with CJ antioxidants or without CJ (control) given to 10 normal individuals after an overnight fast. Plasma antioxidant capacity, glucose, triglycerides, and ascorbate were measured 6 times over 7 h after the consumption of a single 240 mL serving of the two different beverages. The control HFCS caused a slight decrease in plasma antioxidant capacity at all time points and thus an oxidative stress in spite of the presence of ascorbate. CJ produced an increase in plasma antioxidant capacity that was significantly greater than control HFCS at all time points. Postprandial triglycerides, due to fructose in the beverages, were mainly responsible for the oxidative stress and were significantly correlated with the oxidative stress as measured by the antioxidant capacity. Cranberries are an excellent source of high quality antioxidants and should be examined in human supplementation studies.